

DIWA — Digital Interface Web Assistant

Executive Brief | Confidential

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Cavite State University

MEMORANDUM

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OFFICE: Office of the VP for Administrative and Support Services

FROM: DIWA Development Team

DATE: May 2026

SUBJECT: Executive Brief — DIWA, Digital Interface Web Assistant (Current Status & Strategic Value)

1. Overview

DIWA is an artificial intelligence-powered virtual assistant developed specifically for Cavite State University. It is designed to answer common student and staff inquiries automatically, accurately, and instantly — 24 hours a day, 7 days a week.

The system consists of two integrated components: DIWA AI, the Python-based artificial intelligence and API backend; and DIWA Web, a modern web-based chat interface. Together, they form a complete, production-ready solution built entirely by the Cvsu development team using open-source technology at zero licensing cost.

As of May 2026, DIWA is fully operational and production-ready, having completed all core development, testing, and documentation phases.

2. Current Capabilities

DIWA currently provides the following capabilities:

- **Instant Student Q&A:** Responds to student questions covering 23 inquiry categories including admissions requirements, tuition and fees, enrollment procedures, scholarship programs, and courses offered — in under 75 milliseconds per query.
- **24/7 Availability:** Operates continuously without manual intervention, ensuring students have access to accurate information at any time, including outside office hours and during peak enrollment periods.
- **Automated Chat Logging:** All conversations are automatically recorded in a secure SQLite database with daily JSON backups, providing an auditable record and enabling data analysis.
- **Real-Time Analytics Dashboard:** An administrative dashboard displays live metrics on query volume, topic distribution, response times, and usage trends — giving management actionable insight into student information needs.

- **Web Chat Interface (DIWA Web):** A clean, modern, mobile-responsive chat interface accessible from any browser or device, requiring no app installation by students.
- **REST API Integration:** A documented application programming interface (API) with 15+ endpoints enables integration with existing CvSU web platforms and student portals.
- **Docker-Based Deployment:** The entire system can be launched on university servers with a single command, ensuring consistent and reliable deployment across environments.

3. Performance Metrics

The following metrics reflect DIWA's current performance against industry-standard benchmarks for AI-powered chatbot systems:

Metric	Value	Significance
Answer Accuracy	95.59%	Exceeds typical industry threshold of 90% for production chatbots
Response Time	< 75 ms	Near-instantaneous — students receive answers faster than a page load
Training Data Coverage	227 patterns	Covers real CvSU student inquiry scenarios across 23 topic categories
System Uptime	100%	Fully containerised; designed for uninterrupted 24/7 operation
Deployment Size	79 KB model	Lightweight AI model; minimal impact on server resources

4. Strategic Value to Cavite State University

DIWA delivers direct and measurable value across four strategic dimensions:

Operational Efficiency. By automating responses to high-volume, repetitive student inquiries, DIWA frees administrative and academic support staff to focus on complex, high-value tasks. This reduces workload pressure during peak enrollment and admission periods.

Student Experience. Students receive immediate, consistent, and accurate answers regardless of the time of day or staffing availability. This directly supports student satisfaction, reduces confusion during critical decision-making periods, and reinforces the university's commitment to accessible, quality service.

Zero Recurring Cost. DIWA is built entirely on open-source technology by the in-house development team. There are no software licensing fees, vendor subscriptions, or ongoing third-party costs. The only requirement is standard university server infrastructure.

Data-Driven Administration. Every student query is automatically logged and categorised. This gives management unprecedented visibility into what students most frequently ask, enabling evidence-based decisions about information resources, staffing, and communication strategies.

5. Current Development Status

PRODUCT ON READY

All core features are complete, tested, and documented. DIWA is ready for university-wide deployment and is currently awaiting formal administrative endorsement.

Completed Components:

- Naive Bayes AI model — 95.59% accuracy, 8-second training time
- REST API backend — 15+ endpoints with Swagger documentation
- Automated chat logging — SQLite database with daily JSON backups
- Real-time analytics dashboard — live metrics and query trends
- DIWA Web chat interface — responsive, browser-based, mobile-friendly
- Docker containerisation — one-command production deployment
- Full technical documentation — setup, training, and API guides

Planned Enhancements (Next Phase):

- Neural Network integration — enhanced AI for complex or ambiguous queries
- Expanded topic coverage — growing from 23 to 50+ intent categories
- Admin management portal — faculty-managed responses without developer access
- Filipino language support — serving a broader range of students
- Mobile application — Progressive Web App for student smartphones

6. Recommended Actions

The development team respectfully recommends the following actions from the Office of the Vice President for Administrative and Support Services:

1. Endorse University-Wide Deployment

Grant formal approval to deploy DIWA as CvSU's official virtual assistant. This will enable the team to publish the system on the university website, student portal, and official communication channels.

2. Designate an IT Operations Owner

Assign the ICTO or relevant IT unit as the operational custodian of DIWA, responsible for server uptime, system updates, and ongoing model improvements as student inquiry patterns evolve.

3. Commission Expanded Coverage

Authorise the development team to expand DIWA's knowledge base to cover all colleges, departments, and administrative offices — ensuring every student query across CvSU is addressed accurately.